

Supplementary Material

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1.1 Datasets

We compiled a high-quality benchmark dataset for NP-target affinity prediction.

The data preprocessing steps were as follows:

Data Collection and Integration: The data foundation of this study originates from three authoritative public databases: NPASS[?], CMAUP[?], and InflamNat[?]. We systematically collected raw natural product-target interaction data, ensuring each data unit contained the core information essential for subsequent analysis. This includes the SMILES strings of natural product ligands, the amino acid sequences of protein targets (corresponding to UniProt IDs), the experimentally determined affinity types (Ki, Kd), and their corresponding specific numerical values. This step provides a broad and authoritative source of raw data for the entire dataset construction.

Data Preprocessing and Cleaning: Following the acquisition of raw data, we executed a rigorous data cleaning and standardization pipeline to ensure data quality. Initially, invalid entries with affinity values beyond physically reasonable ranges or those severely lacking key information (e.g., SMILES, sequence) were removed. Subsequently, redundant entries from different databases that represented the same natural product-target interaction pair were consolidated, ensuring the uniqueness of each interaction pair within the dataset. This process established a clean, unbiased data foundation for subsequent model training and evaluation.

Ligand and Receptor 3D Structure Preparation: To provide the predictive model with crucial three-dimensional spatial structural information that determines bioactivity, we generated predictive three-dimensional complex conformations for each interaction pair. The initial 3D structures of ligands were generated from their SMILES strings using the RDKit toolkit and optimized via energy minimization. For protein receptors, the preparation of their 3D structures followed two distinct pathways depending on the availability of suitable experimental co-crystallized structures. When an experimental co-crystal structure was available, we screened it based on the 2D topological similarity between its co-crystallized ligand and the natural product ligand under current study, and extracted the geometric parameters of the binding pocket from it. When a suitable experimental structure was unavailable, we employed experimentally

resolved high-quality structures (resolution $< 2 \text{ \AA}$) or AlphaFold2-predicted protein structures, used the fpocket software to predict potential binding pockets, and selected 1 to 3 most promising candidate pockets based on druggability scores from which geometric parameters were extracted.

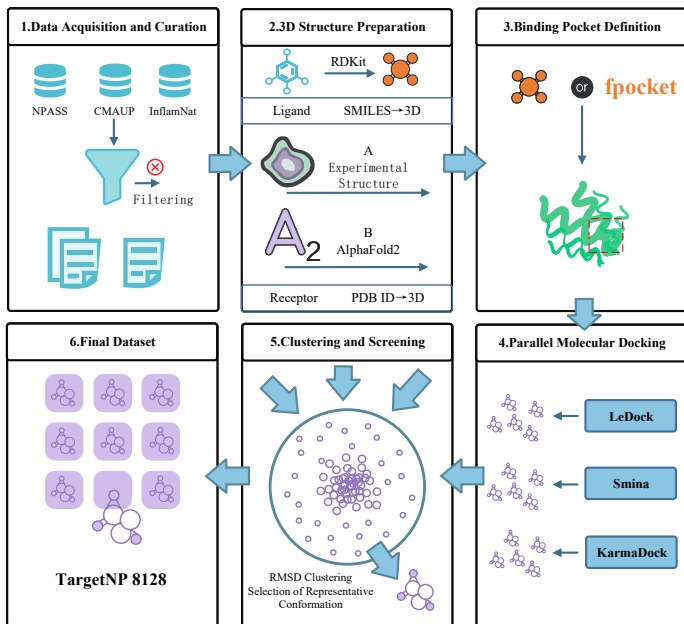


Fig. 1. The construction process of the TargetNP-8128 dataset.

Parallel Docking and Representative Conformation Selection: To comprehensively explore binding modes and reduce reliance on any single algorithm, we employed three molecular docking programs—LeDock[?], Smina[?], and KarmaDock[?—for parallel docking calculations. The docking search space was strictly defined according to the pocket geometric parameters extracted in the preceding steps, with each method generating up to 30 binding conformations per predicted pocket. After docking produced a large number of candidate conformations, we adopted a structure-based clustering and filtering strategy to select the most stable and representative conformation. All conformations were first merged and subjected to cluster analysis based on RMSD. The primary cluster containing the largest number of conformations was then selected. Within this cluster, the average RMSD between each conformation and all other conformations in the cluster (denoted as 3DScore) was calculated. Finally, the conformation with the smallest average RMSD—i.e., the geometric center conformation of the primary cluster—was selected as the final 3D structural representative for

that interaction pair. This method avoids bias from relying solely on docking scores and ensures a consensus on the structural reasonableness of the selected conformation.

Through the systematic process described above, we ultimately obtained a dataset containing 8,128 high-quality NP-target complexes, naming it TargetNP-8128. The construction of this dataset lays a solid foundation for developing predictive models capable of effectively handling the complex structural characteristics of natural products.

Table 1. Basic Statistical Information of the Datasets

Dataset	Type	Ligand No.	Protein No.	Affinity No.	Affinity Type
Davis	Sequence-based	68	442	30056	Kd
KIBA	Sequence-based	229	2111	118254	Kd, Ki, IC50
PDBBind2020	Structure-based	19443	19443	19443	Kd, Ki, IC50
TargetNP-8128	Seq & Struct	2112	1588	8128	Kd, Ki

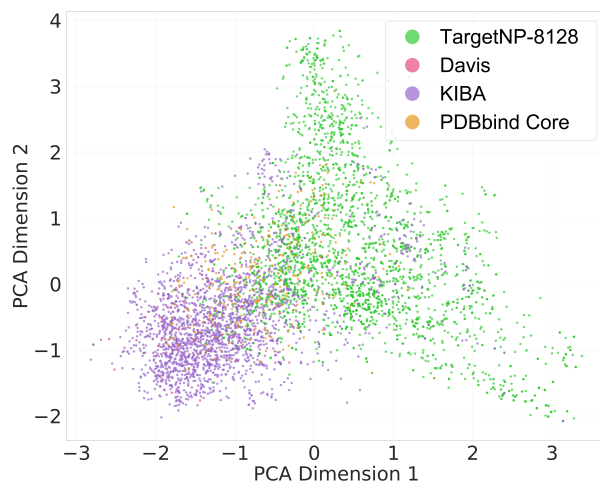


Fig. 2. PCA dimensionality reduction visualization of the dataset.

1.2 Implementation Details

To ensure the full reproducibility of the TargetNPBind model, this supplementary material details its architecture, parameters, and training process. The model overall follows an encoder-fusion-predictor paradigm, with the encoder

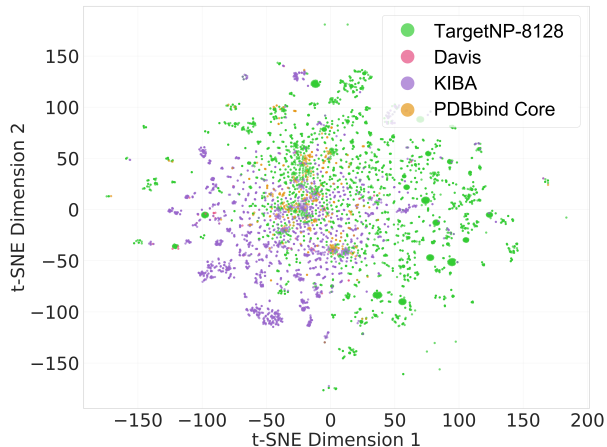


Fig. 3. t-SNE dimensionality reduction visualization of the dataset.

consisting of three modules. The molecular graph encoder takes 79-dimensional atomic features as input, which include 78-dimensional basic chemical features (such as atom types, bond types) and a 1-dimensional crucial pharmacophore indicator bit. This indicator bit is generated by predefined 7 classes of SMARTS patterns (the complete list is shown in Supplementary Table S1), used to mark key pharmacophore features at the atomic level, such as hydrogen bond donors, hydrogen bond acceptors, hydrophobic chains, aromatic rings, etc., thereby explicitly guiding the model to focus on chemical regions relevant to bioactivity. The features are then extracted through three layers of GraphSAGE convolutional layers: the first layer is a pharmacophore-enhanced layer, using max aggregation and ReLU activation, which increases the dimension to 256; the second and third layers are standard GraphSAGE layers, expanding the dimension to 512 and finally refining to 128, respectively, and then obtaining a 128-dimensional molecular representation z_{mol} through global pooling.

Table 2. Definitions of the seven pharmacophore SMARTS patterns used for atomic-level feature enhancement.

Pharmacophore Feature	SMARTS Pattern
Hydrogen Bond Donor (HBD)	<chem>[O,N;!H0][H]</chem>
Hydrogen Bond Acceptor (HBA)	<chem>[C,c](=[O])[O,N;H0]</chem>
Hydrophobic Alkyl Chain	<chem>[C;H2,H1][C;H2,H1]</chem>
Aromatic Ring System	<chem>a1aaaaa1</chem>
Carboxylic Acid	<chem>C(=O)[O;H1][H]</chem>
Hydroxyl Group	<chem>[O;H1][H]</chem>
Amino Group	<chem>[N;H2,H1][H]</chem>

The protein sequence encoder embeds the amino acid sequence into a 128-dimensional vector, then extracts hierarchical features through three 1D convolutional layers (each with kernel size 8, padding 4, and 32 channels) and max-pooling layers with a stride of 2, and finally projects it into a 128-dimensional protein representation z_{prot} .

The 3D interaction graph encoder constructs a heterogeneous graph containing residue and atom nodes, and utilizes a single-layer Equivariant Graph Neural Network (EGNN) to process spatial geometric information. Among them, the message, coordinate update, and node update networks are all MLPs with a hidden layer dimension of 256, using SiLU activation, and supplemented with Layer Normalization and Dropout to stabilize training. Finally, a 256-dimensional graph representation z_{3D} is obtained through average pooling.

The feature fusion module first concatenates z_{mol} and z_{prot} to obtain a 256-dimensional 2D feature z_{2D} , and then dynamically fuses it with z_{3D} through a gating mechanism to obtain the final 256-dimensional feature z_{final} . The prediction head consists of two fully connected layers (with dimensions 512 and 256, respectively, both using ReLU activation and Dropout), and finally outputs the predicted affinity value. The training objective is to minimize the mean squared error.

The model uses the Adam optimizer (learning rate 5e-4, weight decay 1e-5) with a batch size of 64, and employs an early stopping strategy based on the validation set MAE. The complete implementation code, hyperparameter configurations, and preprocessing scripts have been publicly released on a GitHub repository (link in the main text) for reproduction and verification.

1.3 Detailed results for the ligand cold-start scenario

Table 3. Performance comparison under the protein cold partitioning scenario.

	Methods	RMSE	MSE	MAE	CI	Spearman	Pearson
Sequence	AttentionDTA	1.595	2.547	1.172	0.610	0.316	0.320
	DEAttentionDTA	1.609	2.590	1.205	0.585	0.246	0.273
	MREDTA	1.702	2.892	1.275	0.575	0.223	0.259
	DeepGLSTM	1.672	2.803	1.219	0.603	0.319	0.317
Graph	GraphDTA						
	GINConvNet	1.663	2.777	1.256	0.594	0.275	0.324
	GATNet	<u>1.530</u>	<u>2.342</u>	<u>1.136</u>	0.604	0.302	<u>0.335</u>
	GAT_GCN	1.537	2.365	1.168	<u>0.612</u>	0.323	0.333
	GCNNet	1.544	2.385	1.161	0.606	0.307	0.329
	MDCT-DTA	1.596	2.551	1.187	0.592	0.266	0.291
Structure	FAST						
	3D-CNN	1.715	2.944	1.259	0.592	0.271	0.249
	SG-CNN	1.687	2.884	1.231	0.577	0.227	0.253
	fusion	1.533	2.412	1.167	0.588	0.230	0.288
	LGN	1.698	2.890	1.270	0.565	0.203	0.255
	TargetNPBind	1.499	2.247	1.130	0.617	<u>0.320</u>	0.351

1.4 Baselines

Sequence-based methods.

AttentionDTA takes drug SMILES and protein amino acid sequences as input, using separate 1D-CNNs to extract their local features. Its key innovation is a bidirectional multi-head attention mechanism that automatically focuses on sequence segments critical for binding affinity, enhancing model interpretability while maintaining the efficiency of sequence-based approaches.

DEAttentionDTA processes both the global protein sequence and the sequence of its local binding pocket, along with the ligand SMILES. It employs a dynamic embedding layer and a self-attention mechanism to effectively correlate contextual information around the binding site, leading to enhanced feature fusion.

MREDTA also employs sequence inputs but adopts a pre-trained BERT model and Transformer architecture as its encoder. Leveraging large-scale biomolecular corpora, BERT provides rich prior knowledge, while the Transformer captures long-range dependencies, enabling the model to learn deeper and more generalizable molecular semantic representations. **DeepGLSTM** targets drug-target affinity prediction for repurposing. Its core combines a GCN—augmented with higher-order adjacency matrices to capture multi-hop topological information—with a bidirectional LSTM for encoding protein sequences.

Two-dimensional molecular graph-based methods.

GraphDTA pioneered the use of 2D molecular graphs—converted from SMILES strings—and graph neural networks (GNNs) to learn drug representations. Through rigorous experiments, it demonstrated the superiority of graph-based over sequence-based representations for capturing molecular structures, establishing a foundation for subsequent research.

MDCT-DTA also processes drug molecular graphs and protein sequences. It introduces a multi-scale graph diffusion convolution module to capture complex topological interactions within the molecule, and employs a local inter-layer interaction mechanism to enable fine-grained, dynamic feature fusion between drug and protein representations, moving beyond simple concatenation.

Three-dimensional structure-based methods.

LGN processes 3D protein-ligand complexes as graphs. To address the severe node imbalance between proteins and ligands, it separately extracts features from the complex, the ligand subgraph, and interaction fingerprints, then uses ensemble learning to combine them, improving model balance and robustness.

FAST improves protein-ligand affinity prediction by fusing 3D structural information. It processes the 3D complex in parallel: a 3D-CNN extracts voxel grid features, and a Spatial Graph CNN (SG-CNN) captures interatomic graph features, combining the complementary strengths of both approaches for robust predictions.

1.5 Sensitivity Analysis of the Model to Docking Conformations

To test the model’s sensitivity to specific docking conformations, we compared training with randomly selected conformations versus the consensus geometric-

center conformation. Performance exhibited minimal difference across all evaluation scenarios (Tables S3–S5), indicating the model learns inherent interaction patterns robust to conformational variation rather than overfitting to any single pose.

Table 4. Performance comparison under the pair cold partitioning scenario

Methods	RMSE	MSE	MAE	CI	Spearman	Pearson
TargetNPBind-random	1.213	1.413	0.852	0.749	0.666	0.646
TargetNPBind-Top1	1.197	1.434	0.838	0.759	0.6770	0.660

Table 5. Performance comparison under the ligand cold partitioning scenario

Methods	RMSE	MSE	MAE	CI	Spearman	Pearson
TargetNPBind-random	1.493	2.209	1.124	0.607	0.3090	0.352
TargetNPBind-Top1	1.499	2.247	1.130	0.617	0.320	0.351

Table 6. Performance comparison under the protein cold partitioning scenario

Methods	RMSE	MSE	MAE	CI	Spearman	Pearson
TargetNPBind-random	1.674	2.773	1.341	0.642	0.463	0.487
TargetNPBind-Top1	1.662	2.770	1.340	0.667	0.482	0.505